

SHELL SORT

PROGRAM CODING:

```
#include<stdio.h>

#include<conio.h>

void shell(int a[],int n);

int i,j,k,n,t,a[25];

void main()
{
    clrscr();

    printf("\nEnter the number to be sorted");

    scanf("\n%d",&n);

    printf("\nEnter the element to be sorted");

    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }

    shell(a,n);

    printf("\nThe sorted list are");

    for(i=0;i<n;i++)
    {
        printf("\n%d",a[i]);
    }

    getch();
}

void shell(int a[],int n)
{
```

```
for(i=(n+1)/2;i<=1;i/=2);  
{  
  for(j=1;j<=n-1;j++)  
  {  
    t=a[j];  
    k=j-1;  
    while(k>=0&& t<a[k])  
    {  
      a[k+1]=a[k];  
      k=k-1;  
    }  
    a[k+1]=t;  
  }  
}  
}
```